

Eren Dogan

Last updated: August 2, 2026

PhD Candidate in Computer Science and Engineering
Baskin School of Engineering
University of California, Santa Cruz

erdogan@ucsc.edu
<https://edogan.us>
<https://linked.in/in/erendn>

EDUCATION

PhD in Computer Science and Engineering

GPA: 3.86/4.00

Advisor: Prof. Matthew Guthaus

University of California, Santa Cruz

September 2022 – June 2027 (expected)

BS in Computer Science

GPA: 4.00/4.00

Honors: Valedictorian, Full-tuition scholarship

Ozyegin University

September 2017 – June 2022

WORK EXPERIENCE

Student Researcher

Sunnyvale, CA

Google

April 2026 – Present

RESEARCH

Publications

C.3 **Eren Dogan** and Matthew R. Guthaus

“Effective Capacitance Modeling Using Graph Neural Networks.”

Paper at *27th International Symposium on Quality Electronic Design (ISQED 2026)*, San Francisco, CA, USA, April 2026.

C.2 Bugra Onal, **Eren Dogan**, Muhammad Hadir Khan, and Matthew R. Guthaus

“GAT-Steiner: Rectilinear Steiner Minimal Tree Prediction.”

Paper at *4^{3rd}* IEEE/ACM International Conference on Computer-Aided Design (ICCAD 2024), Newark, NJ, USA, October 2024.

C.1 Muhammad Hadir Khan, Bugra Onal, **Eren Dogan**, and Matthew R. Guthaus

“VLSI Hypergraph Partitioning with Deep Learning”

Work-in-Progress Paper at *6th ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD 2024)*, Snowbird, UT, USA, September 2024.

P.1 **Eren Dogan**, H. Fatih Ugurdag, and Hasan Unlu

“Deep Compression for PyTorch Model Deployment on Microcontrollers.”

Poster at *35th International Conference on VLSI Design and 21st International Conference on Embedded Systems (VLSID 2022)*, Bengaluru, Karnataka, India, February 2022.

W.1 **Eren Dogan**, H. Fatih Ugurdag, and Matthew R. Guthaus

“OpenCache: An Open-Source OpenRAM Based Cache Generator.”

Paper at *4th Workshop on Open-Source EDA Technology (WOSET 2021)*, Zoom, November 2021.

Contributions

- 2025- [OpenROAD](#): Contributed to OpenROAD, an open-source RTL-to-GDS flow. Implemented global gate sizing for the timing repair stage.
- 2022-24 [OpenRAM](#): Contributed to OpenRAM, an open-source SRAM compiler. Implemented a new graph-based, BVH-accelerated supply and signal escape router from scratch.
- 2021 [OpenCache](#): Developer of OpenCache, an open-source custom cache generator.

TEACHING EXPERIENCE

Teaching Assistant

CSE 101, CSE 102, CSE 127A

University of California, Santa Cruz

September 2022 – March 2026

Teaching Assistant

CS 101, EE 203, CS 240

Ozyegin University

September 2019 – June 2020

SKILLS & EXPERTISE

Languages: C/C++, (System)Verilog, Chisel, Tcl, Python

Frameworks: PyTorch, PyTorch Geometric, Tensorflow, Spektral

Miscellaneous: GDSII, SPICE, Linux, Git, Shell, Makefile

Domain Expertise: VLSI, EDA, CAD, RTL Design, Computer Architecture

SERVICE

Reviewer

2025 [62nd ACM/IEEE Design Automation Conference \(DAC 2025\)](#).

2024 [32nd IFIP/IEEE International Conference on Very Large Scale Integration \(VLSI-SoC 2024\)](#).

2024 [6th ACM/IEEE International Symposium on Machine Learning for CAD \(MLCAD 2024\)](#).

Volunteer

2025 [7th ACM/IEEE International Symposium on Machine Learning for CAD \(MLCAD 2025\)](#).

AWARDS

2026 [CITRIS Costas John Spanos Semiconductor Manufacturing Award](#)